

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:
A positive Integer n
Output:
Print the value of the counter variable
Answer:

```
1 #include<stdio.h>
2 int main(){
3     int n;
4     int count=0;
5     scanf("%d",&n);
6     int c= 0;
7     count++;
8     for(int i=n/2; i<n; i++){
9         count++;
10        for(int j=1; j<n; j = 2 * j){
11            count++;
12            for(int k=1; k<n; k = k * 2){
13                count++;
14                c++;
15                count++;
16            }
17            count++;
18        }
19        count++;
20    }
21    count++;
22    printf("%d",count);
23    return 0;
24 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct
Marks for this submission: 1.00/1.00.